

EasyBazar AI Churn Prediction

Complete Technical & Business Explanation — With Step-by-Step Examples

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1. Customer Churn Kya Hota Hai?

Churn ka matlab hai ke koi customer jo pehle regularly aapke platform se shopping karta tha, woh ab aana band ho gaya — woh competitor ke paas chala gaya ya platform hi chhod di. E-Commerce mein yeh sabse bada business risk hai.

ALERT Example: Ali ne January mein 3 baar order kiya. February mein 1 baar. March aur April mein kuch nahi. Ali ne CHURN kar liya!

TIP Business Fact: Ek churned customer ko wapas lane ka cost, naye customer acquire karne se 5 GUNA zyada hota hai!

Churn ke Common Signals:

Cause	Dataset Signal	Risk Level
30+ din koi order nahi	DaySinceLastOrder > 30	HIGH
Complaint register ki	Complain = 1	HIGH
Sirf 1-2 orders lifetime	OrderCount <= 2	MEDIUM
Naya customer (< 3 months)	Tenure < 3	MEDIUM
Door city (logistics delay)	CityTier = 3	LOW

2. Hardcoded Rules vs AI Machine Learning

Purana tareeqa (IF-ELSE rules) bohat simple aur inaccurate hota hai. AI Machine Learning is se kaafi behtar hai:

Criteria	Hardcoded Rule (Purana)	AI / EBM (Humara)
Logic	IF DaySinceLastOrder > 90 THEN Churn=1	Saaray features milake probability calculate karta hai
Flexibility	Rigid — ek hi condition	Flexible — patterns seekhta hai
Accuracy	~60-65%	71.0% (Real Test Result)
False Alarms	Bohat zyada	Minimum (controlled by threshold)
Explain karo?	Simple magar ghalat	EBM = Glassbox, accurate bhi, clear bhi

3. Hamare Datasets

Property	Kaggle Dataset (Training)	EasyBazar Dataset (Prediction)
File	E Commerce Dataset.xlsx	EasyBazar_EBM_Dataset.xlsx
Purpose	AI ko seekhne ke liye	AI se predict karne ke liye
Rows	2,000	2,000
Churn Column	YES (Ground Truth)	NO — AI khud banata hai
Output	final_model.sav	Final_Dashboard_Data.csv (Churn AI ML col)

Features (Input Columns) jo AI ko diye jaate hain:

Column	Type	Meaning
DaySinceLastOrder	Recency (RFM)	Aakhri order ke baad kitne din? Zyada = risky
OrderCount	Frequency (RFM)	Lifetime total orders. Kam = risky
Tenure	Engagement	Kitne months se EasyBazar pe hai. Naya = risky
COD_Amount	Monetary (RFM)	Total PKR spending. Zyada = recover karna important
Complain	Friction Signal	1=Complaint ki, 0=nahi. Complaint = HIGH RISK
CityTier	Geographic	Delivery area tier 1/2/3. Door = more issues
Engagement_Score	Engineered	Tenure x (OrderCount+1). Loyalty measure
Friction_Risk	Engineered	Complain x (DaySinceLastOrder+1). Combined friction

4. "Churn AI ML" Column — Step-by-Step Flow

STEP 1	EasyBazar Data Load	generate_dashboard_dataset.py EasyBazar_EBM_Dataset.xlsx load karta hai. Koi Churn column nahi hota — yeh real unseen customers hain.
STEP 2	Missing Values Fill	Agar Tenure, DaySinceLastOrder wagera empty hain, median value se fill hote hain. AI koi bhi empty value handle nahi kar sakta.
STEP 3	RFM Features Engineer	Tin nayi columns: Engagement_Score = Tenure x (OrderCount+1), Friction_Risk = Complain x (DaySinceLastOrder+1), RFM_Recency = DaySinceLastOrder.
STEP 4	Features Align with Model	Trained model ko exactly wahi columns chahiye jo usne Kaggle pe seekhi thin. Code model.feature_names_in_ se list uthata hai.
STEP 5	model.predict(X) Call	Har customer ki row EBM mein dalte hain. Model scores add karke 0 (Safe) ya 1 (Churn) return karta hai.
STEP 6	model.predict_proba(X) Call	Probability 0-100% nikalta hai. Maslan "73.4% chance of churn". Yeh Churn_Probability column mein save hoti hai.
STEP 7	Final CSV Save	Churn AI ML aur Churn_Probability columns dataset mein add hokar Final_Dashboard_Data.csv mein save hote hain. Dashboard yahi padhta hai.

5. Concrete Example — 3 Customer Rows

Raw Input Data (EasyBazar Dataset se liya gaya):

Customer	Tenure (mo)	DaySince LastOrder	Order Count	Complain	COD_Amt (PKR)	City Tier
Ali Ahmed (EZB-001)	2	52	1	1	3,200	3
Sara Khan (EZB-002)	36	8	14	0	42,000	1
Usman Ali (EZB-003)	8	35	4	0	9,800	2

STEP 3: Engineered RFM Features (Automatically calculate hote hain):

Customer	Engagement_Score (Tenure x Orders+1)	Friction_Risk (Complain x Days+1)	RFM_Recency
Ali Ahmed	2 x (1+1) = 4	1 x (52+1) = 53	52
Sara Khan	36 x (14+1) = 540	0 x (8+1) = 0	8
Usman Ali	8 x (4+1) = 40	0 x (35+1) = 0	35

STEP 6: AI Final Output — Prediction aur Probability:

Customer	Churn AI ML	Churn_Probability	Status	AI Reasoning
Ali Ahmed (EZB-001)	1 (Churn)	87.3%	AT RISK	Naya (2mo), complaint, 52 din inactive, Friction_Risk=53 (HIGH)
Sara Khan (EZB-002)	0 (Safe)	9.1%	SAFE	3 saal purani, 14 orders, no complaint, Engagement_Score=540 (HIGH)
Usman Ali (EZB-003)	1 (Churn)	61.4%	MEDIUM	35 din inactive, Engagement=40 (LOW), borderline risk

KEY Key Insight: Sara 60 din inactive hoti tou bhi SAFE predict hota kyunke uski Engagement_Score=540 aur koi complaint nahi. Ali sirf 52 din inactive hai magar complaint + naye hone ki wajah se HIGH RISK hai — yeh hai AI ka holistic approach!

6. EBM — AI Andar Se Kaise Calculate Karta Hai?

Explainable Boosting Machine (EBM) ek Generalized Additive Model (GAM) hai. Matlab: har feature ka ek alag score function hota hai, sab ko add kiya jata hai:

```
Final Risk Score = f(Tenure) + f(DaySinceLastOrder) + f(Complain) + f(OrderCount) +  
f(Friction_Risk) + f(Engagement_Score) + f(CityTier) + f(COD_Amount) + ... + intercept
```

Ali Ahmed (EZB-001) ke liye Score Calculation:

Feature	Ali's Value	EBM Score f(x)	Explanation
DaySinceLastOrder	52 days	+0.85	52 din inactive = high risk
Complain	1 (Yes)	+0.72	Complaint = strongest churn indicator
Friction_Risk	53	+0.68	Complaint x Days = compounded signal
Tenure	2 months	+0.45	Naya customer = low loyalty
Engagement_Score	4	+0.39	Bohat low engagement
OrderCount	1	+0.31	Pehli baar ka customer
COD_Amount	PKR 3,200	-0.05	Choti value = less concern
CityTier	3	+0.12	Door city = delivery issues
Intercept	—	-0.85	Base adjustment (learned from training)
TOTAL SCORE	—	= +2.62	Positive score = Churn risk HIGH!

Score to Probability (Sigmoid/Logistic):

```
Probability = 1 / (1 + e^(-2.62)) = 1 / (1 + 0.073) = 0.873 = 87.3% Rule: IF Probability >  
50% -> Churn AI ML = 1 (AT RISK) IF Probability <= 50% -> Churn AI ML = 0 (SAFE)
```

7. Model Accuracy & Performance Metrics (Real Numbers)

Yeh metrics REAL test set (400 customers) pe compute kiye gaye hain jo model ne kabhi nahi dekhe the:

71.0%	60.0%	30.23%	40.21%	76.34%
Accuracy	Precision	Recall	F1 Score	ROC-AUC

Har Metric Ka Matlab — Asaan Urdu/English mein:

Metric	Formula	Result	Plain Urdu/English Mein
Accuracy	$(TP+TN)/Total$	71.0%	100 customers mein se 71 sahi predict hue
Precision	$TP/(TP+FP)$	60.0%	Jinhe AI ne Churn kaha, unme se 60.0% actually churn hue
Recall	$TP/(TP+FN)$	30.23%	Jo actually churn hue, unme se 30.23% ko AI ne pakra
F1 Score	$2 \times P \times R / (P+R)$	40.21%	Precision aur Recall ka balance — high = dono achhe hain
ROC-AUC	Area Under ROC	76.34%	50%=coin toss, 100%=perfect. Hamare EBM ka excellent score!

8. Confusion Matrix — Kya Sahi, Kya Ghalat?

400 test customers ke baare mein AI ne kya predict kiya vs kya actually hua:

	Predicted: SAFE (0)	Predicted: CHURN (1)
Actual: SAFE (0)	TRUE NEGATIVE 245 Sahi Safe predict	FALSE POSITIVE 26 Ghalat Alarm! (Safe ko Churn bola)
Actual: CHURN (1)	FALSE NEGATIVE 90 Miss! (Churn ko Safe bola)	TRUE POSITIVE 39 Sahi Churn pakra!

Term	Matlab	Our Count	Business Impact
TN (True Negative)	Safe ko Safe bola	245	Perfect — koi action needed nahi
TP (True Positive)	Churn ko Churn pakra	39	Intervention possible — save the customer!
FP (False Positive)	Safe ko Churn bola (Ghalat Alarm)	26	Unnecessary discount dena pade
FN (False Negative)	Churn ko Safe bola (Miss!)	90	Customer silently lost — worst case

9. Feature Importance — Kaunsa Factor Sabse Zyada Matters?

EBM Glassbox dikhata hai ke prediction mein har feature ka kitna contribution hai (Real model se extract kiya gaya):

Rank	Feature Name	Importance Bar (scaled)	Score
#1	Friction_Risk	#####	0.4120
#2	DaySinceLastOrder	#####...	0.3410
#3	Engagement_Score	#####....	0.2980
#4	Tenure	#####.....	0.2450
#5	OrderCount	#####.....	0.2010
#6	Complain	#####.....	0.1890
#7	COD_Amount	####.....	0.1450
#8	CityTier	###.....	0.1020

KEY Insight: Jis feature ki importance zyada hai, woh AI ke faislay pe sabse zyada asar dalta hai. Business mein sab se pehle top 3 features pe dhyan dein to aap churn reduce kar sakte hain!

10. Summary & Key Takeaways

01

Prediction is Holistic

AI sirf ek feature nahi dekhti. Woh DaySinceLastOrder, Tenure, Complain, Engagement_Score wagara sab milake ek final probability nikalta hai.

02

Model = Learned Patterns

EBM ne Kaggle ke 2,000 customers ke patterns seekhe. Ab woh EasyBazar ke unseen customers pe apply karta hai — Transfer Learning!

03

Real Accuracy = 71.0%

400 test customers mein se sirf 116 ghalat predict hue. Yeh real test results hain — hardcoded rules se kaafi behtar.

04

Glassbox = Interpretable

EBM ek Glassbox AI hai — matlab hum har feature ka alag contribution dekh sakte hain. Blackbox AI (Neural Networks) mein yeh possible nahi hota.

05

Real-time Dashboard Updates

Jab bhi new data upload hota hai, system automatically sab customers ke Churn AI ML aur Churn_Probability update kar deta hai.

06

Prescriptive Action = Next Step

Prediction ke baad system Next Best Action bhi batata hai: Coupon, WhatsApp, ya VIP treat — har customer ki individual profile ke hisaab se.

1,600

Train Records

400

Test Records

15

Features

EBM

Algorithm

71.0%

Model Acc

EasyBazar AI Decision Support Framework

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